Time Synchronization

The Qbit Timing Synchronization Solution is a high-precision synchronization platform designed to deliver deterministic timing, phase alignment, and accurate frequency distribution across distributed network infrastructures. Built using advanced synchronization technologies including IEEE 1588-2008 Precision Time Protocol (PTP), Synchronous Ethernet (SyncE), and hardware timestamping, the platform enables sub-nanosecond timing accuracy over standard optical fiber networks.

Introduction

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The solution is optimized for mission-critical applications requiring reliable time and frequency synchronization, low jitter performance, deterministic latency, and scalable network operation. It is ideal for telecom infrastructure, scientific research facilities, smart grids, aerospace and defense systems, industrial automation, broadcasting, and financial trading environments.
By integrating advanced clock management, high-speed Ethernet switching, and robust management capabilities, the Timing Synchronization Solution ensures stable and precise operation in highly demanding deployments.

Architecture

The Timing Synchronization Solution architecture combines a high-performance FPGA processing subsystem with a deterministic Ethernet switching core and precision timing engine.
The system integrates:

- Precision timing engine implementing IEEE 1588 PTP and SyncE
- Hardware timestamping modules
- Deterministic Layer-2 Ethernet switching
- High-stability clock generation and distribution
- Optical fiber communication interfaces
- Embedded management and monitoring subsystem
- Low-jitter PLL and oscillator network
- Secure remote configuration and diagnostics

This architecture enables precise timing transfer, deterministic data communication, and scalable synchronization deployment over long-distance optical networks.

Features

- **Deterministic Timing Synchronization:** The system delivers highly accurate synchronization with sub-nanosecond precision using hardware timestamping and advanced clock recovery mechanisms. It supports precise phase, frequency, and time alignment across distributed network nodes.
- **IEEE 1588-2008 PTP Support:** Implements Precision Time Protocol (PTPv2) with advanced timing extensions to ensure accurate synchronization over Ethernet networks.
- **Synchronous Ethernet (SyncE):** Provides stable frequency transfer and clock recovery for highly synchronized network environments.
- **Low-Latency Data Distribution:** Supports deterministic packet forwarding and low-latency communication suitable for real-time and time-sensitive applications.
- **Flexible Optical Connectivity:** Supports multi-port optical fiber interfaces for long-distance timing and data distribution.
- **High-Stability Clocking Architecture:** Integrated PLLs, oscillators, and clock distribution networks ensure low jitter and reliable holdover performance.
- **Advanced Management Interface:** Secure management via SSH, SNMP, web interface, and UART for configuration, diagnostics, and monitoring.
- **Redundant & Reliable Operation:** Designed for continuous operation with robust power architecture and environmental monitoring.

Specification

SYNCHRONIZATION & PERFORMANCE	
Parameter	**Specification**
Time Synchronization Accuracy	• < 500 ps typical, < 1 ns guaranteed
Jitter Performance	• Picosecond-level
Data Delivery	• Fully deterministic
Packet Forwarding	• Deterministic low-latency switching
Maximum Fiber Distance	• Up to 100 km
Fiber Calibration	• Dynamic asymmetry calibration
Timestamping	• Hardware-based
Holdover Stability	• High-stability oscillator support
NETWORK INTERFACES	
Total Optical Ports	• 24 × SFP/SFP+
Active Ethernet Ports	• 20 × 1 Gb SFP
Reserved Ports	• 4 × SFP
Management SFP Port	• 1 × 1 Gb SFP

Specification Cont...

RJ45 Ethernet Port	• 1 × 10/100/1000 Mbps
UART Interface	• RS-232 over RJ45
USB Interfaces	• USB-C (UART), USB-A
TIMING & CLOCK INTERFACES	
PPS Input	• LVCMOS-3.3V / 5V TTL compatible
PPS Output	• TTL-5V standard PPS
10 MHz Input	• Sine or digital input
10 MHz Output	• AC-coupled 10 MHz
Reference Clock I/O	• 10 MHz / 62.5 MHz
Auxiliary Timing Port	• AUX (abscal)
PROCESSING RESOURCES	
SoC / FPGA	• Xilinx Zynq Ultrascale+ XCZU17EG
Application Processor	• Quad-core ARM Cortex-A53 up to 1.5 GHz
Real-time Processor	• Dual-core ARM Cortex-R5 up to 600 MHz
High-Speed Transceivers	• 32 GTH (16.3 Gb/s), 16 GTY (32.75 Gb/s)
CLOCKING RESOURCES	
Main Oscillator	• 25 MHz VCTCXO
DAC	• AD5662 16-bit
Auxiliary Oscillators	• 2 × Si549
Grandmaster PLL	• LMX2594
Main Loop PLL	• HMC7044

Specification Cont...

MEMORY & STORAGE	
System Memory	4 GB SO-DIMM
eMMC Storage	4 GB
NOR Flash	Dual Quad-SPI (2 × 1 Gb)
EEPROM	I²C Serial EEPROM 64K
Expansion Storage	M.2 SATA SSD Support
TIMING PROTOCOL SUPPORT	
Timing Protocol	IEEE 1588-2008 PTP
SyncE Support	Supported
Grandmaster Mode	Supported
Boundary Clock Mode	Supported
PPS Distribution	Supported
Event Timestamping	Hardware-based precise tagging
NETWORKING & PROTOCOL SUPPORT	
Switching Standards	IEEE 802.1x
VLAN Support	IEEE 802.1Q
Multicast	IGMP Snooping
Loop Prevention	STP / RSTP
IP Stack	TCP/IP, DHCP, ARP, DNS, TFTP
Remote Access	SSH, SNMP v1/v2c
Time Backup	NTP Support
MANAGEMENT & CONFIGURATION	
Web Interface	Browser-based management

Specification Cont...

CLI Access	RS-232 / USB UART
SNMP Monitoring	Supported
Firmware Upgrade	Field upgradable
Diagnostics	Real-time monitoring & logging
USER INTERFACE & INDICATORS	
Display	OLED
Status LEDs	Programmable red/green
Buttons	Reset + 2 control buttons
Rear Panel Indicator	Location LED
MECHANICAL SPECIFICATIONS	
Form Factor	1U Rack Mount
Height	44.45 mm
Width	482.6 mm
Depth	310 mm
ENVIRONMENTAL SPECIFICATIONS	
Operating Temperature	0°C to +45°C
Storage Temperature	-25°C to +60°C
Humidity	10% to 90% non-condensing

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